Task 4 — Business Sales Analytics (Optional Capstone)

Alfido Tech Data Science Internship | Project deliverable

Task 4: Business Sales and Customer Analytics

Dataset and methodology

• Source: `data/raw/business_sales_customers.csv`.

• Records: 30,000; columns: 41.

• Duplicate full rows: 0.

• Missing cells across the dataset: 0.

• Revenue metric: sum of `Order_Amount`, grouped by the dataset's Year and Month fields.

• Profit metric: sum of `Profit_Amount`, grouped by `Product_Category`.

• Currency and units were not independently verified; monetary figures are reported in source units.

Findings

• Total recorded order amount: 11,370,043.99.

• Average recorded order amount: 379.00.

• Median recorded order amount: 171.03.

• Total recorded profit amount: 2,215,916.27.

• Monthly revenue series contains 48 month groups with valid year/month fields.

• Unique customers according to `Customer_ID`: 8,683.

• Highest recorded revenue category: Electronics (6,320,059.31).

• Highest recorded total profit category: Electronics (1,040,688.00).

Business recommendations

• Monitor monthly revenue and profit separately; do not use discount amount as a proxy for revenue.

• Compare category revenue with category profit before allocating promotions.

• Validate currency, units, return handling, order statuses and duplicate order identifiers before operational decisions.

• Treat the results as descriptive analytics, not proof that a category or promotion caused better performance.

• Validate any forecast on a chronological holdout period before using it for planning.

Outputs

• `outputs/task4_monthly_revenue.csv`

• `outputs/task4_monthly_revenue.png`

• `outputs/task4_sales_by_category_summary.csv`

• `outputs/task4_profit_by_category_summary.csv`

• `outputs/task4_revenue_by_category.png`

• `outputs/task4_profit_by_category.png`

Recommendations and next steps

• Track revenue, profit, order volume and average order value over time.

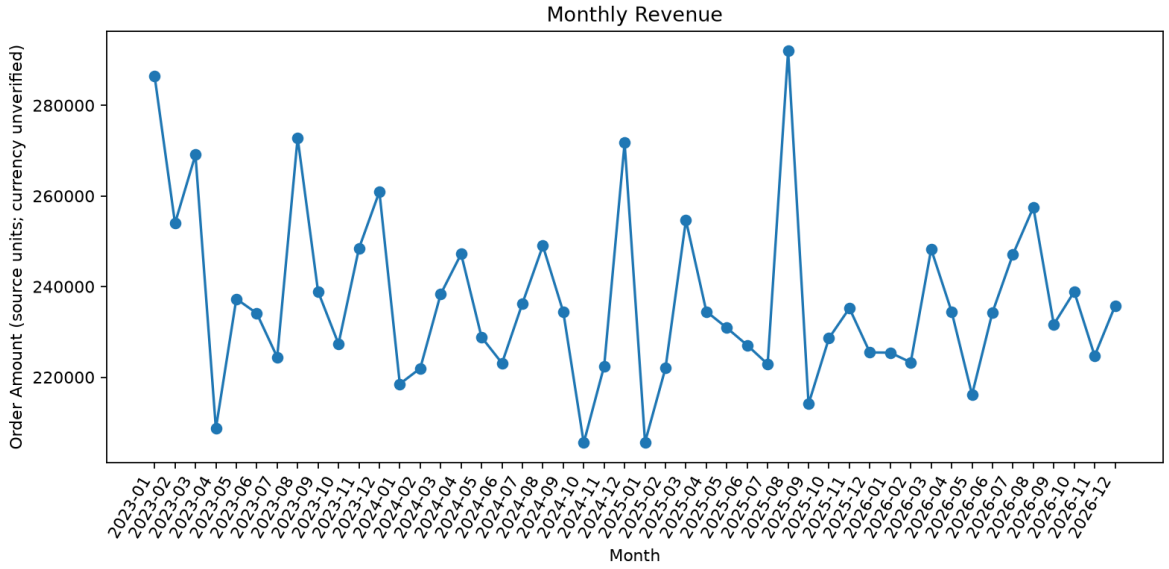
• Compare category-level sales and profit before reallocating promotions.

• Check returns, discounts, missing values, currency and duplicate order IDs.

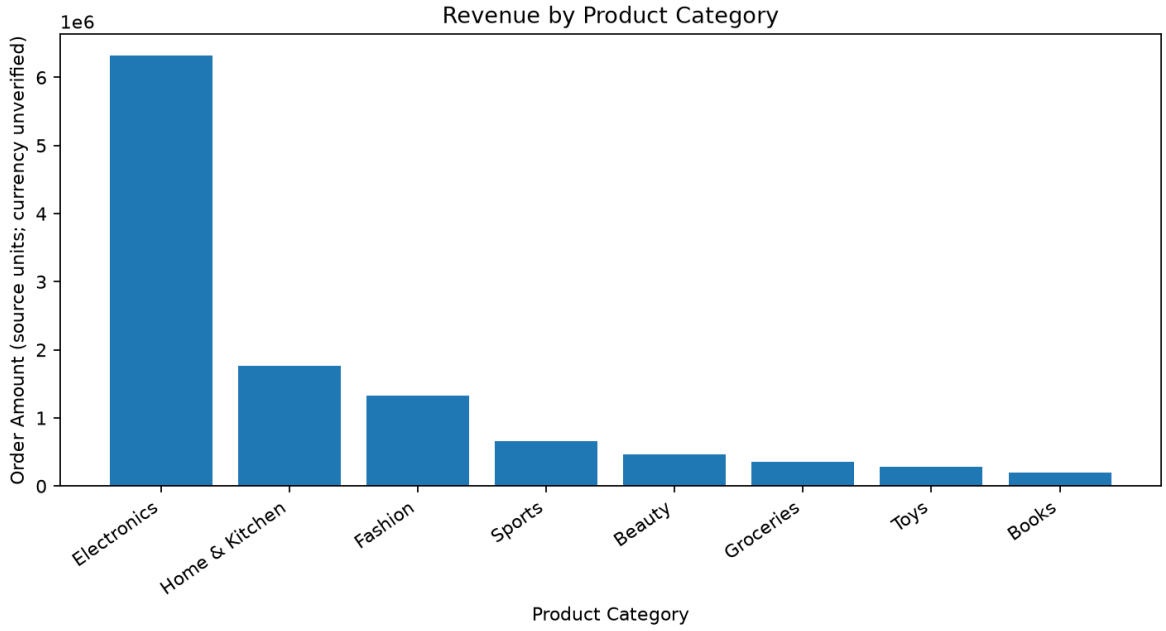
• Segment customers only after confirming the meaning and reliability of customer identifiers.

• Present findings as descriptive analytics; forecast only after a validated time-based evaluation.

Task4 Monthly Revenue



Task4 Revenue By Category



Task4 Profit By Category

